GH-300

Pull Requests

& Code Review with Copilot

PR summaries, AI-assisted review, and the Copilot cloud agent that closes the loop.

PR Summary | AI Code Review | Custom Instructions | Cloud Agent | Protected branches

Sandeep Soni · Deccansoft Software Services Pvt Ltd

Deck 5 · GH-300 Certified GitHub Copilot Expert Training

OVERVIEW

Three Copilot capabilities in the PR workflow

PR Summary

Auto-generated description + title from the diff. Lets reviewers find what changed and why before they open the first file.

AI Code Review

Copilot reviews the PR with full project context. Suggests fixes you can apply with one click. Same engine in IDE and on github.com.

Copilot cloud agent

Assign an issue to Copilot — it researches, codes, runs CI, self-reviews, and opens a draft PR. You drive the merge.

Honest framing

Supplement, not substitute. Copilot Code Review is a second-pair-of-eyes pass; it does not replace human judgment on intent, business correctness, or architecture.

It misses things. Context is bounded by the model window and time budget — large PRs get partial coverage. Pair with human review for production paths.

Speed without quality is debt. Use Copilot to find more — not to merge more.

PR Summary — auto-generated descriptions and titles

1. Open / create PR
2. Click the Copilot ✦ icon
3. Choose Summary
4. Review & publish

What you get

- Prose overview of the change.
- Bulleted list of key changes, linked to the diff lines.
- A generated title from your commits — separate button in the title field.
- Inline regeneration without losing your edits.
- Available on existing PRs as a comment, not only on new ones.

Editorial discipline

- Read every generated summary before merging. Copilot will over-explain trivial changes and under-explain risky ones.
- Add the WHY — Copilot describes the WHAT well; intent is your job.
- Link to the issue / ADR / incident — generated summaries do not.
- Edit before publishing — drafts persist if you regenerate.

AI Code Review — how it works

Full project context

Reads the diff plus related files, cross-file dependencies, the PR description, and commit messages.

Multi-angle analysis

Correctness, security, readability, performance — in any language Copilot supports. Same engine in IDE and github.com.

Suggested fixes

One-click commits for accepted suggestions. Batch multiple suggestions into a single commit.

Auto-fix via Cloud

Hand a review batch to the Copilot cloud agent. It opens a new PR with the fixes — closing the loop.

Review modes · Estimated cost per review

Lite (fast, targeted)

About $0.05 to $1, plus Actions minutes.

Balanced (default, higher-reasoning model)

About $0.25 to $5, plus Actions minutes.

WALKTHROUGH

Code Review — step by step on github.com

1. **Open your PR**
   Create or navigate to an existing pull request.

2. **Request Copilot review**
   Reviewers menu → select Copilot.

3. **Wait**
   Usually < 30 seconds for a small PR.

4. **Read comments**
   Copilot leaves a "Comment" review — never Approve / Request Changes.

5. **Apply suggestions**
   Single fix → commit, or batch multiple → one commit.

6. **Iterate**
   React, reply, or resolve — just like a human review.

Note: Copilot reviews do not count toward required approvals by default — but rulesets can be configured to weigh them in for non-protected branches. Verify your specific branch policy before relying on this behaviour.

Copilot approvals (public preview) let Copilot submit an approving review that satisfies required approvals; the approval is dismissed when new commits are pushed.

Code Review in your IDE — shift left, save the round-trip

VS Code

- Select code → right-click → Generate Code → Review.
- Or ask in Copilot Chat: "review this code" with #selection.
- Comments appear inline + in the Comments panel.
- Honors .github/copilot-instructions.md.

Visual Studio

- Requires Visual Studio 17.14 or later.
- Select code in the editor → request review.
- Same review quality as github.com.
- Custom instructions sourced from head branch.

Review whole PRs from the VS Code Pull Requests panel

- Open the GitHub Pull Requests extension — review the full PR diff without leaving the editor.
- Request a Copilot review from inside the panel — comments stream back into VS Code.
- @copilot in PR comments triggers iteration; commits flow back to the branch automatically.

Automatic Code Review — configure once, review everything

For yourself

Copilot Settings → Automatic Copilot code review → Enabled. Requires a paid Copilot plan.

For a repository

Repo Settings → Rules → Rulesets → branch ruleset → "Automatically request Copilot code review" — now a standalone rule with Run on each push and Run on drafts sub-settings.

For the org

Org Settings → Copilot → Features → enable Automatic Copilot code review. Set review effort level and a default Actions runner org-wide.

⚠ Billing — live since June 1, 2026

What changed: All Copilot plans now bill on GitHub AI Credits, and each code review also consumes GitHub Actions minutes on private repos. Public repos stay free.

Who is billed: Automatic reviews bill the PR author, manual reviews bill the requester, and bot-opened PRs bill the organization.

Why this matters: Auto-review-on-every-push across many repos can spike Actions minute usage. Audit your rulesets and set a budget cap.

Admin lever: Org admins can now set a default Actions runner for code review across all repos — no per-repo setup. That is the practical answer to the cost concern.

Recommendation: Enable review on draft PRs; consider disabling review-on-every-push if your team rebases frequently.

NEW · INSTRUCTIONS

Custom instructions for code review — the highest-leverage quality knob

How it works

The repo or org "Code review instructions" box, plus copilot-instructions.md, *.instructions.md, and AGENTS.md. The 4,000-character cap was removed in June 2026.

What to put there

Only what differs from your code style guide. Be specific about what the reviewer should focus on — and what it should NOT comment on (formatting handled by Prettier / dotnet-format, etc.).

review section

```
# Code review focus for this repo

Project: payments service · runtime: .NET 8

Focus on:
  - Concurrency: new async paths must propagate CancellationToken end-to-end.
  - Money: decimal only — flag any double / float arithmetic on monetary values.
  - Audit: every state transition on Order must emit an AuditEvent before return.
  - PII: do not log raw card numbers; mask to last 4 digits using Mask.LastFour().

Do NOT comment on:
  - Formatting — handled by dotnet-format.
  - Naming on private members — _camelCase is intentional.
```

Copilot cloud agent — from issue to pull request

1

Assign issue

Assignees → Copilot. Optional prompt for guidance.

2

Research + plan

Reads the repo, related files, and the issue body. Drafts an internal plan.

3

Code + test

Edits in an ephemeral env (GitHub Actions). Runs tests + linters, auto-corrects.

4

Draft PR

Opens a PR and requests your review. @copilot in comments to iterate.

5

Merge

You merge. Or push your own commits on the same branch.

Triggers from anywhere — github.com is not the only entry point

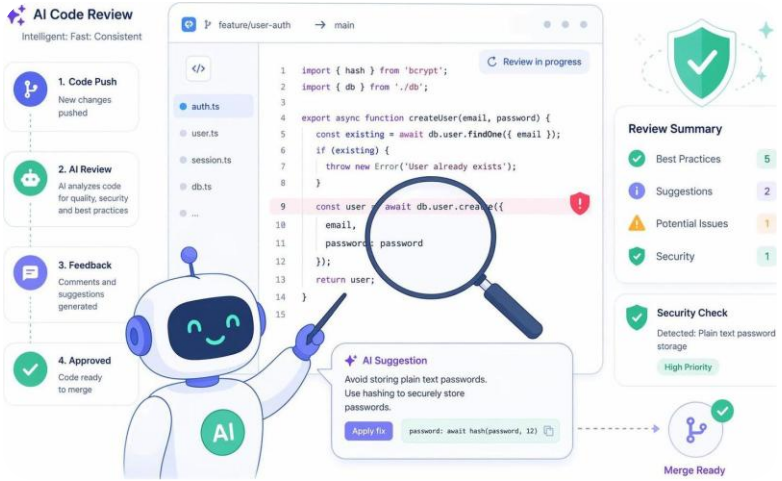
- github.com — Assignees → Copilot on any issue.
- GitHub CLI — gh agent-task create / gh copilot agent for non-interactive dispatch (verify the verb against current docs).
- VS Code · Visual Studio · GitHub Mobile · MCP-enabled IDEs — same agent, surfaced where you work.

Reviewing a Copilot-created PR — five moves

@copilot in comments	Mention @copilot to ask for changes. It adds a 👀 reaction and pushes commits to the branch.
Batch your feedback	Submit review comments together; Copilot responds to comments from anyone with write access.
Choose a model	Use the model picker on github.com when submitting review comments. Default = the model that created the PR. (Updated Apr 2026.)
Push your own changes	Check out the branch and edit yourself. Mix human and AI work on one PR.
Request a separate PR	Ask Copilot in plain language to open a new PR for fixes instead of pushing here.

Cloud Agent self-review + security scan — closing the loop

1. **Draft**
   Agent writes code and tests in the ephemeral env.
2. **Self-critique**
   Re-reads the diff as a reviewer would. Looks for the issues a human reviewer would raise first.
3. **Security scan**
   Runs CodeQL / dependency review on the diff. Flags injection, hardcoded secrets, vulnerable packages.
4. **Self-fix**
   Applies the resulting suggestions automatically — same engine as Copilot Code Review.
5. **Draft PR**
   PR opened with a self-summary and the list of issues already fixed. Human review starts higher up the stack.



What this changes for human reviewers

The first round of trivial findings (lint, obvious null-derefs, common security smells) is already cleared before the PR opens. Your job moves up the stack: intent, business correctness, architectural fit. Read the agent's self-summary — it tells you which issues were already addressed.

What Copilot checks — and what it cannot

Correctness

Logic errors, off-by-one, null refs, unhandled edges.

Security

Injection, unsafe data handling, auth gaps, hardcoded secrets.

Readability

Naming, complexity, dead code, magic numbers.

Performance

Inefficient loops, N+1 queries, missing memoisation.

Style & standards

Project conventions from copilot-instructions.md and AGENTS.md.

Cross-file impact

How the diff affects callers; breaking changes flagged.

Limitations to remember

- Not guaranteed to catch all issues — always pair with human review.
- Large PRs get partial coverage — bounded by model window and time budget.
- Cannot replace domain expertise — business correctness is still yours.
- Will not catch issues outside the diff unless explicitly referenced.

Protected branches + required reviewers — how Copilot fits

Default behaviour

- Copilot reviews are "Comment" reviews — never Approve / Request Changes.
- They never block merging on protected branches.
- They do not count toward required approvals.
- Your own approval on a Copilot-created PR does not count if you requested the agent.

Rulesets can do more

- Require Copilot review on every push to a ruleset-matched branch.
- Run review on draft PRs to catch errors before human review starts.
- Some experimental rulesets allow weighing Copilot review as N-of-required for non-prod branches — verify before relying on it.

Practical policy for most teams

Production branches (main, release/*): 1+ human approval required, Copilot review enabled but advisory.

Feature branches: Copilot review on draft PRs (cheap signal, catches issues before human time).

Bot PRs (dependabot, renovate): Auto-review on every push; merge only after green CI + Copilot has no error-severity comments.

Cloud-Agent-created PRs: Always require at least one human approval — self-review is supplement, not replacement.

CLOSE

Key takeaways

1

Three capabilities, one workflow

PR Summary, AI Code Review, Copilot cloud agent — same project context, same custom instructions.

2

Specific review instructions win

The review instruction is the highest-leverage knob for review quality. Tell Copilot what to focus on AND what to ignore.

3

Shift review left

Review in the IDE (and the VS Code PR panel) before opening the PR — cheaper signal, faster loop.

4

Cloud agent closes the loop

Self-review + security scan before the PR opens. Human review starts higher up the stack.

5

Mind the June 2026 billing change

Auto-review-on-every-push moves to GitHub Actions minutes. Audit rulesets and set budgets before cutover.

6

Copilot supplements humans

It will not catch everything. Always pair with a human reviewer on production paths.

Thank you.

Bring your messiest PR — we will review it together.

Sandeep Soni · Deccansoft Software Services Pvt Ltd